

ALI HASSAN

Software Engineer | Full-Stack Developer | Data Scientist | AI/ML Engineer

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PROFESSIONAL SUMMARY

Motivated Computer Software Engineering graduate with hands-on experience building full-stack web applications and AI/ML systems. Proven ability to deliver end-to-end solutions — from data pipelines and predictive models to production-ready REST APIs and React frontends. Seeking a software engineering or data science role where I can contribute technical skills and a strong drive to learn.

EDUCATION

Lahore Garrison University (LGU)

Bachelor of Science in Computer Software Engineering

2022 – 2026

Lahore, Pakistan

Cumulative GPA 3.31 | Specialization in AI/ML, Data Science, and Software Engineering

Forman Christian College (FCCU)

Intermediate in Computer Science (ICS)

2020 – 2022

Lahore, Pakistan

PROJECTS

ImmunoAI: Multimodal AI Medical Diagnostic System — *Django, React, PyTorch, LangChain, Pinecone*

- Built a hybrid multimodal AI system fusing clinical biomarkers with MRI neuroimaging, achieving 88% validation accuracy via ResNet50 CNN and a stacking ensemble (Random Forest, XGBoost, LightGBM) with weighted fusion
- Integrated Explainable AI (SHAP plots, Grad-CAM heatmaps) and a RAG-powered medical chatbot using LangChain and Pinecone Vector DB for evidence-based query responses
- Delivered a full-stack Django REST API + React application featuring a patient portal, specialist dashboard, appointment management, and secure PDF report generation

Grand Voyage — Hotel Management Platform — *Node.js, Express.js, EJS, MySQL, MVC Architecture*

- Developed a full-stack hotel booking platform using MVC architecture with server-side rendering via EJS and a scalable MySQL schema supporting room management, reservations, and dynamic content delivery
- Implemented structured controller-based request handling and reusable view templates demonstrating enterprise-level code organization

ML-Driven Prediction of Falcon 9 Landing Success — *Python, Scikit-learn, Plotly Dash, SQL*

- Built an end-to-end ML pipeline covering SpaceX API integration, web scraping, SQL-based EDA, and feature engineering; benchmarked Logistic Regression, SVM, Decision Trees, and KNN for highest accuracy
- Created an interactive Plotly Dash dashboard with geospatial mapping and real-time analytics for launch site performance, payload relationships, and success rate visualization

TECHNICAL SKILLS

Data Science: Pandas, NumPy, Matplotlib, Power BI, Scikit-learn, PyTorch

Web Development: React, Node.js, Django, FastAPI, Express.js, HTML5, CSS3, Bootstrap, Tailwind CSS

Languages: Python, C++, JavaScript

Databases: MySQL, PostgreSQL, SQLite, MongoDB, Pinecone Vector DB

Tools & Platforms: Git, REST APIs, LangChain, SHAP, Plotly Dash, Jupyter Notebook

CERTIFICATIONS

- IBM Data Science Specialization — Comprehensive training in data science methodologies and tools
- Web Development Udemy Bootcamp — Full-stack web development with modern frameworks
- Google AI Essentials — Foundational knowledge in artificial intelligence and machine learning
- Generative AI: Elevate Your Data Science Career — Advanced techniques in generative AI models
- Microsoft Office 365 — Proficiency in Microsoft productivity and collaboration tools